

Forecasting Methods

Box-Jenkins methodology.

Forecasting with ARIMA models.

Introduction to Seasonal ARIMA models.

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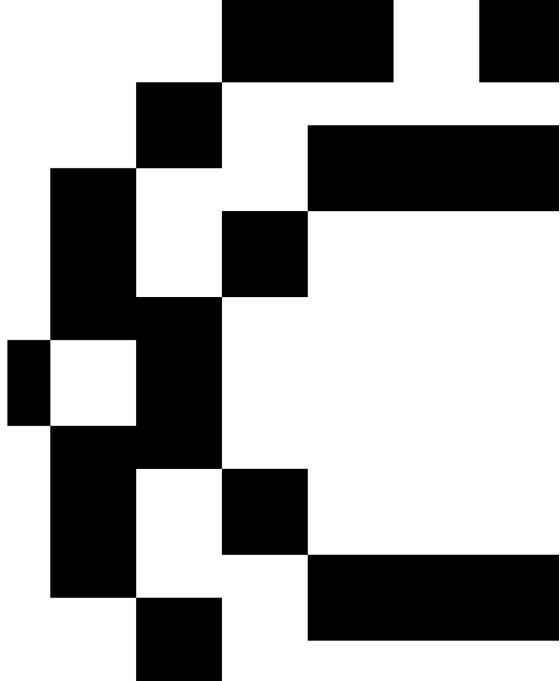
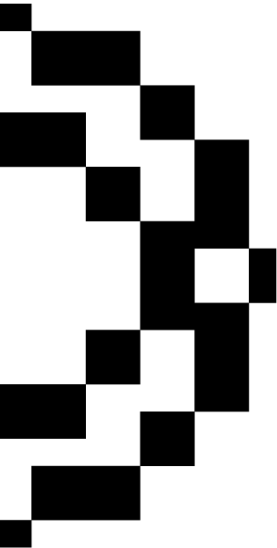


Table of contents

1. Box-Jenkins methodology
2. Forecasting with ARIMA models
3. Introduction to SARIMA Models



Box-Jenkins methodology

Introduction to the Box-Jenkins methodology

- The objective of this group of slides is to present the typical set of tools used to identify the appropriate ARIMA model for a given time series data and to estimate the parameters of the model.
- Furthermore, we will discuss some indispensable tools to validate the model.
- A widely used methodology for model building is the Box-Jenkins methodology.
- The Box-Jenkins methodology uses an iterative cycle based on visual and statistical tools obtained from the time series data.

Iterative cycle for model building

Three steps procedure

- **Tentative identification** - use all relevant information from the data to suggest some candidate models to be analysed in more detail during the next stages.
- **Estimation** - use the available data to estimate the model and test statistical hypotheses regarding the parameters.
- **Diagnostic checking** - use a set of tools to verify if the model is satisfactory or if it is inadequate and should be replaced by another model.

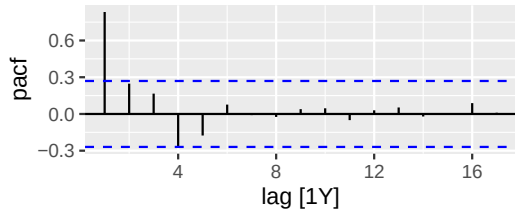
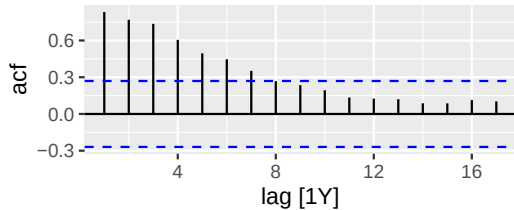
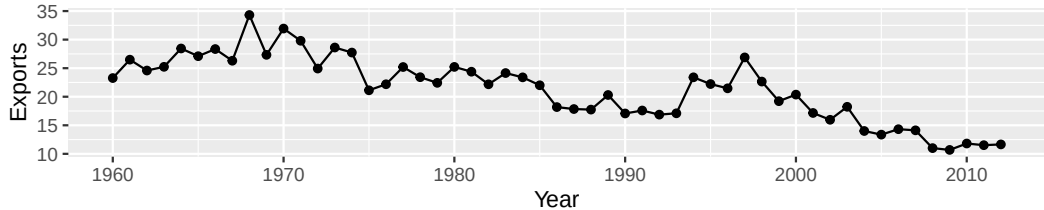
First Step: Identification

- In the first step of the tentative identification stage, one should take differences as many times as necessary until the series of interest, y_t , becomes stationary.
- In general, either the time series is stationary (possibly after removing the linear deterministic trend or applying some variance transformation) or 1, at most, 2 differences are necessary to stationarize the series.
- To evaluate the need of this transformation, the following tools are topically used:
 - ⇒ Visual inspection of the graph in levels/first differences
 - ⇒ Visual inspection of the correlogram
 - ⇒ Formal statistical inference methods (unit root/stationarity tests)

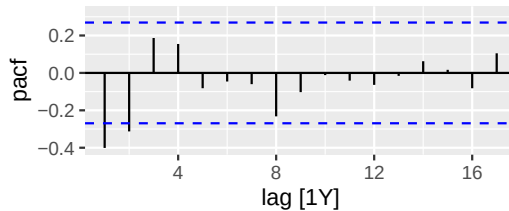
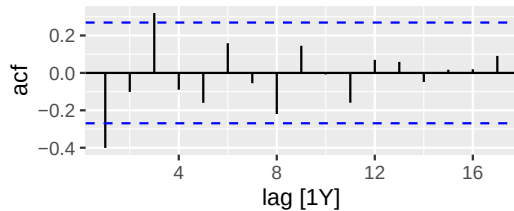
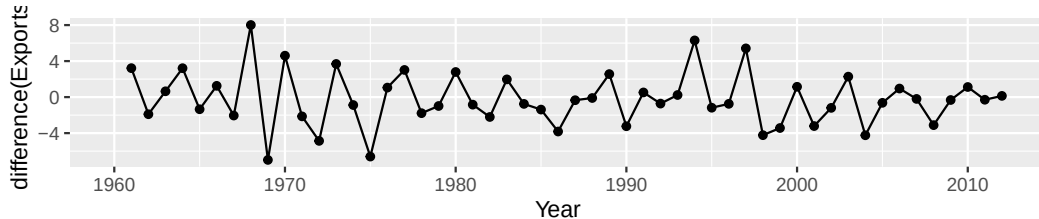
Stationarity?

```
global_economy %>%  
  filter(Code == "CAF") %>%  
  gg_tsddisplay(Exports, plot_type='partial')
```

Stationarity: What do you conclude?



Stationarity?



Identification – Choice of the orders p and q of the ARMA

- Having stationarized the time series with the appropriate transformations, the next step is to identify the orders of the autoregressive polynomial (p) and moving average polynomial (q).
- For this purpose, recall the behavior of the ACF and PACF of the class of ARMA models:

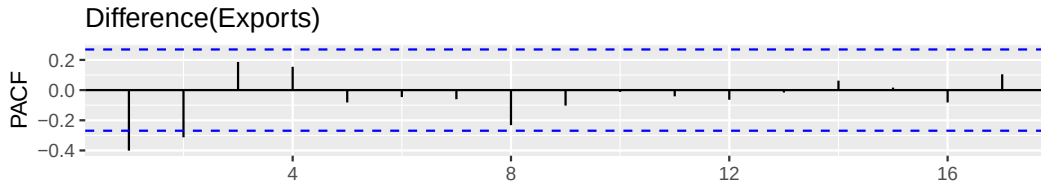
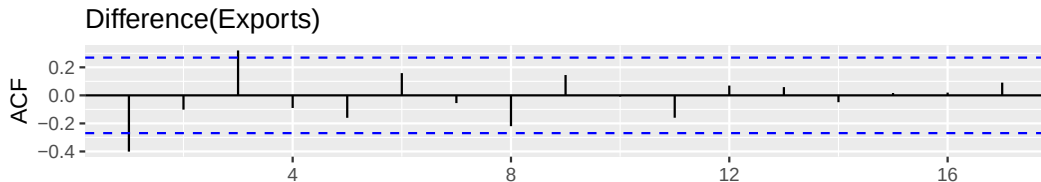
Model	ACF
AR(p)	<u>Infinite</u> and tails off as exponential decay or damped sine wave
MA(q)	<u>Finite</u> and cuts off after lag q
ARMA(p,q)	<u>Infinite</u> and tails off as exponential decay or damped sine wave after lag q

Identification – Choice of the orders p and q of the ARMA

Model	PACF
AR(p)	<u>Finite</u> and cuts off after lag p
MA(q)	<u>Infinite</u> and tails off as exponential decay or damped sine wave
ARMA(p,q)	<u>Infinite</u> and tails off as exponential decay or damped sine wave after lag p

- Obviously, in practice we don't know the theoretical ACF and PACF of the process that generated the data. We only have access to sample estimates of the ACF and PACF.
- We then choose the orders q and p that match as close as possible the patterns of the sample ACF and PACF with the theoretical ones.

Is the series stationary? Identify candidate models.



Is the series stationary? Identify candidate models.

- The PACF is suggestive of an $AR(2)$ model; so an initial candidate model is an $ARIMA(2,1,0)$.
- The ACF suggest an $MA(3)$ model, so an alternative candidate is an $ARIMA(0,1,3)$.
- Since the third lag of the PACF is close to the significance level, one possible candidate model could be an $AR(3)$.
- We can also use the `ARIMA()` function for automated model selection.

Tentative Identification – Information criteria

- Given the limitations associated with the subjective use of the sample ACF and PACF, a good deal of research effort was devoted to device more automatic and straightforward model selection criteria.
- One of the most successful procedures is the use of the so called information criteria.
- In general, the information criteria are written as:

$$IC = \log \hat{\sigma}_e^2 + \text{penalty}(\#parameters)$$

- Our objective is to MINIMIZE the information criteria!
- The most popular information criteria have different penalty functions and, consequently, different properties:

Tentative Identification – Information criteria

1. Akaike Information Criteria (AIC):

$$AIC = \log \hat{\sigma}_e^2 + \frac{2 \cdot k}{T}$$

2. Hannan-Quinn Information Criteria (HQ):

$$HQ = \log \hat{\sigma}_e^2 + \frac{2 \cdot k \cdot \log(\log(T))}{T}$$

3. Bayesian or Schwarz Information Criteria (BIC ou SIC):

$$BIC = \log \hat{\sigma}_e^2 + \frac{k \cdot \log(T)}{T}$$

where k is the number of estimated coefficients. For the ARIMA models we have $k = p + q + 1$.

Tentative Identification – Information criteria

- The HC e BIC almost certainly choose the true model for **large** samples but select overparametrized models in small samples
- AIC performs better than the BIC and HQ in **small** samples but is asymptotically biased and overparametrized models in large samples.
- Sometimes the Information Criteria do not reach a consensus. Indeed, it is perfectly plausible that each Information Criteria defends a different model specification.
- The Information Criteria should only be used as a supplementary tool in the model building process. They should **never replace** the careful analysis of the ACF and PACF and the residual diagnostics.

Tentative Identification – Information criteria

IMPORTANT: the Information Criteria should be applied whenever we are in doubt about some adequate specifications for the time series of interest.

- Why? Because the Information Criteria are only reliable if the model is correctly specified.
- In other words, if the candidate models have problems, the Information Criteria are **not valid**.

Tentative Identification – Information criteria

```
caf_fit <- global_economy %>%
  filter(Code == "CAF") %>%
  filter(Year <= 2012) %>%
  model(arima210 = ARIMA(Exports ~ pdq(2,1,0)),
        arima013 = ARIMA(Exports ~ pdq(0,1,3)),
        arima310 = ARIMA(Exports ~ pdq(3,1,0)))

caf_fit %>% pivot_longer(!Country, names_to = "Model name",
                        values_to = "Orders")

## # A mable: 3 x 3
## # Key:      Country, Model name [3]
##   Country      `Model name`      Orders
##   <fct>         <chr>           <model>
## 1 Central African Republic arima210 <ARIMA(2,1,0)>
## 2 Central African Republic arima013 <ARIMA(0,1,3)>
## 3 Central African Republic arima310 <ARIMA(3,1,0)>

glance(caf_fit) %>%
  arrange(AICc) %>%
  select(.model:BIC)

## # A tibble: 3 x 6
##   .model  sigma2 log_lik  AIC  AICc  BIC
##   <chr>    <dbl>  <dbl> <dbl> <dbl> <dbl>
## 1 arima013  6.97   -123.  254.  255.  262.
## 2 arima210  7.21   -124.  255.  255.  260.
## 3 arima310  7.03   -123.  254.  255.  262.
```

Second Step: Estimation

- Having identified one or more candidate models, the parameters of the model must be estimated. There are various possible estimation methods:
 1. Method of moments
 2. Least squares (ordinary or non linear)
 3. Conditional and Unconditional maximum likelihood
 4. Method of moments (generalized)
 5. State space representation and the Kalman filter.
- However, the most commonly used estimation method is Maximum Likelihood.

Estimation

```
global_economy %>%  
  filter(Code == 'CAF') %>%  
  filter(Year <= 2012) %>%  
  model(ARIMA(Exports ~ pdq(0,1,3))) %>%  
  report()  
  
## Series: Exports  
## Model: ARIMA(0,1,3)  
##  
## Coefficients:  
##          ma1      ma2      ma3  
##      -0.4588  0.0938  0.2845  
## s.e.   0.1367  0.1573  0.1351  
##  
## sigma^2 estimated as 6.966:  log likelihood=-122.99  
## AIC=253.99   AICc=254.84   BIC=261.79
```

Estimation

- If the model is well specified the estimators for $\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q$ are consistent and asymptotically normal distributed.
- Hence, if a candidate model is adequate we can test for the statistical significance of the coefficients.
- Moreover, a estimate **not statistically significant** may then be removed from the model.
- A **General-to-specific procedure** is typically applied: remove the regressor with highest p-value, reestimate and reevaluate the residuals of the model. Repeat this process until all the estimates are statistically significant.

Estimation - Restricted ARIMA(0,1,3)

```
global_economy %>%
  filter(Code == 'CAF') %>%
  filter(Year <= 2012) %>%
  model(ARIMA(Exports ~ 1 + pdq(0,1,3, fixed = list(ma2 = 0)))) %>%
  report()

## Series: Exports
## Model: ARIMA(0,1,3) w/ drift
##
## Coefficients:
##          ma1  ma2      ma3  constant
##        -0.4330    0  0.3059   -0.2196
## s.e.    0.1134    0  0.1238    0.3070
##
## sigma^2 estimated as 6.956:  log likelihood=-122.93
## AIC=253.87   AICc=254.72   BIC=261.67
```

Third Step: Diagnostic Checking

- Having identified and estimated a candidate model, the next step is to assess the model adequacy.
- A model is adequate if it satisfies the model assumptions. The most important one is that the error term, ε_t , is a white noise process.
- Recall that the error term, ε_t , is a white noise process if the following assumptions are satisfied:

$$E(\varepsilon_t) = 0$$

$$\text{Var}(\varepsilon_t) = E(\varepsilon_t^2) = \sigma_\varepsilon^2$$

$$\text{Corr}(\varepsilon_t, \varepsilon_{t-k}) = 0, \quad k = 1, 2, \dots$$

Third Step: Diagnostic Checking

- It is possible to prove that if the specified model coincides with the Data Generating Process (DGP), then as $T \rightarrow \infty$ the residuals, $\hat{\varepsilon}_t$, become close to the true errors ε_t .
- Hence, we may expect that $\hat{\varepsilon}_t$ yield valuable evidence concerning the model adequacy.
- To check the zero mean and constant variance assumptions, we may examine the residual plot.
- Furthermore, the ACF of the residuals should be zero at all lags:

$$\rho_k = \text{Corr}(\hat{\varepsilon}_t, \hat{\varepsilon}_{t-k}) = 0, k = 1, 2, \dots$$

- The distribution of estimated autocorrelations is usually approximated by a normal distribution with zero mean and standard error $\frac{1}{\sqrt{T}}$.
- This fact can be used to assess approximately the statistical significance of the autocorrelations.

Third Step: Diagnostic Checking

- However, as shown by Durbin (1970) and Box e Pierce (1970), in many cases the standard error of the residual sample autocorrelations is substantially smaller than than $1/\sqrt{T}$, specially, for low lags.
- Rather than test the statistical significance of each point of the ACF, a more efficient alternative is to test whether, say, the first 30 autocorrelations taken as a whole are statistical significant.
- Ljung-Box statistic: Tests the null of no residual autocorrelation until lag K

$H_0 : \rho_1 = \rho_2 = \dots = \rho_K = 0$ versus the presence of residual autocorrelation

$H_1 : \rho_1 \neq 0 \vee \rho_2 \neq 0 \vee \dots \vee \rho_K \neq 0$ with the test statistic:

$$Q(K) = LB(K) = T \sum_{k=1}^K \hat{\rho}_k^2 \stackrel{a}{\sim} \chi_{(K-p-q)}^2$$

Third Step: Diagnostic Checking

- However, according to Ljung e Box (1978) the Chi-square distribution may not be good approximation to the distribution of the Q statistic under H_0 and a modified version was proposed – the Ljung-Box-Pierce statistics or adjusted Q-statistics.

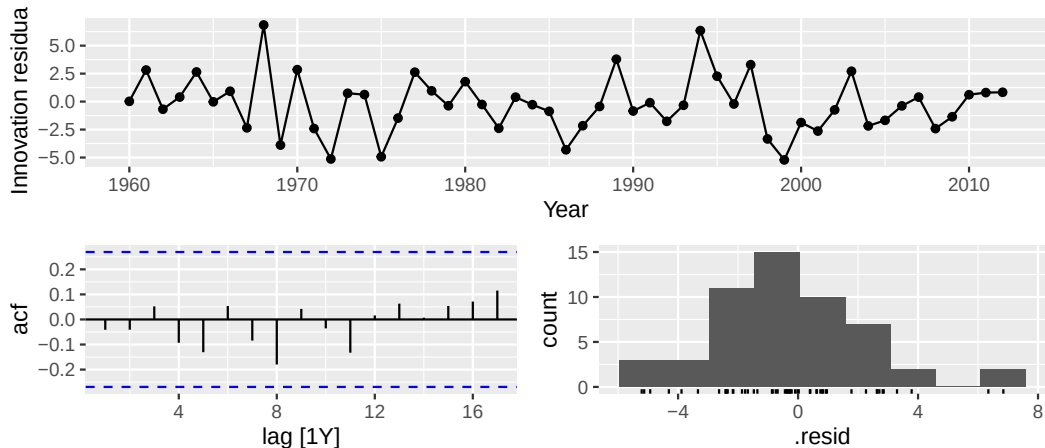
$$LBP(K) = \tilde{Q}(K) = T(T+2) \sum_{k=1}^K \frac{\hat{\rho}_k^2}{T-k} \stackrel{a}{\sim} \chi_{(K-p-q)}^2$$

- If the null hypothesis is rejected, the model is **inadequate** and the model building iterated cycle has to be repeated starting from the **model identification**.
- The choice of K is somewhat arbitrary: a very high K means that a good deal of power may be lost whereas a very low K may ignore important information at higher lags. A popular rule of thumb is to set $K = \sqrt{T}$ and apply the test to values in its neighborhood.

Residuals

```
caf_fit %>%  
  select(arima310) %>%  
  gg_tsresiduals()
```

Residuals

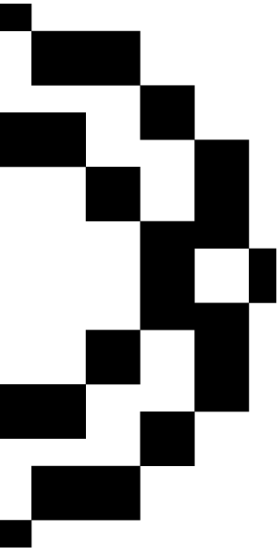


Ljung-Box

```
augment(caf_fit) %>%  
  filter(.model=='arima013') %>%  
  features(.innov, ljung_box, lag = 10, dof = 3)  
  
## # A tibble: 1 x 4  
##   Country      .model  lb_stat lb_pvalue  
##   <fct>      <chr>    <dbl>    <dbl>  
## 1 Central African Republic arima013    4.83    0.681
```

Ljung-Box-Pierce

```
augment(caf_fit) %>%  
  filter(.model=='arima013') %>%  
  features(.innov, box_pierce, lag = 10, dof = 3)  
  
## # A tibble: 1 x 4  
##   Country      .model  bp_stat bp_pvalue  
##   <fct>      <chr>    <dbl>    <dbl>  
## 1 Central African Republic arima013    4.09    0.770
```



Forecasting with ARIMA models

Introduction

- A main goal of building an ARIMA model is to predict the future path of the time series of interest.
- Advantages of using ARIMA models for forecasting:
 - ARIMA models usually perform quite well and often outperform more complicated models.
 - Predictions are simple to obtain after model building as they correspond to linear combinations of present and past values of the time series.
- Disadvantages of using ARIMA models for forecasting:
 - We cannot provide any economic insight in model predictions. In particular it is not possible to attribute the forecast movements and forecast errors to variations in different economic factors/variables.
 - In the same line of thought, it is not possible to build forecasts under alternative macroeconomic scenarios.

Introduction

- In any forecasting exercise, we suppose that we are at time T and we are interested in predicting the value of a time series y_t at a future date $T + 1, T + 2, \dots$
- The prediction of $y_{T+s}, s = 1, 2, \dots$ is based on a set of variables observed at period T .
- For example, the forecasts for ARIMA models are based on present and past values of the variable of interest y_t
- Let $y_{T+s|T}$ denote a forecast obtained at date T of the value of the variable y at date $T + s$, i.e., s steps ahead.
- To evaluate the usefulness of the forecast $y_{T+s|T}$ it is convenient to define a function that summarizes how concerned we are if our forecast $y_{T+s|T}$ is distant from the realized value y_{T+s} .

Introduction

- Very convenient results are obtained from assuming a quadratic loss function as a metric. This function is called Mean Squared Error (MSE) and is defined as:

$$MSE(y_{T+S|T}) = E[(\varepsilon_{T+S|T})^2] = E[(y_{T+S} - y_{T+S|T})^2]$$

where $\varepsilon_{T+S|T}$ denotes the error associated with forecast $y_{T+S|T}$, i.e., $\varepsilon_{T+S|T} = y_{T+S} - y_{T+S|T}$

Introduction

- It is possible to show that the best forecast, i.e., the one that minimizes the mean squared error is given by the expectation of y_{T+s} on the σ -algebra $\mathcal{F}_T$

$$y_{T+s|T} = E[y_{T+s} | \mathcal{F}_T] = E_T(y_{T+s})$$

- Therefore the forecast $y_{\{T+s|T\}}$ for the ARIMA models is obtained as the expectation of y_{T+s} conditional on present and past values of y_t
- We will study unbiased forecasting methods which implies that $E(y_{T+s|T}) = y_{T+s}$ and so we have the results:

$$MSE(y_{T+s|T}) = E[(\varepsilon_{T+s|T})^2] = Var(\varepsilon_{T+s|T}) = \sigma_{\varepsilon_{T+s|T}}^2$$

Point Forecasts

Point forecasts can be calculated using the following three steps.

1. Expand the ARIMA equation so that y_t is on the left hand side and all other terms are on the right.
2. Rewrite the equation by replacing y_t with y_{T+S} .
3. On the right hand side of the equation, replace future observations with their forecasts, future errors with zero, and past errors with the corresponding residuals.

Beginning with $S = 1$, these steps are then repeated for $S = 1, 2, \dots$ until all forecasts have been calculated.

Point forecasts

ARIMA(3, 1, 1) forecasts: Step 1

$$(1 - \phi_1 L - \phi_2 L^2 - \phi_3 L^3)(1 - L)y_t = (1 - \theta_1 L)\varepsilon_t$$

$$\begin{aligned} [1 - (1 + \phi_1)L + (\phi_1 - \phi_2)L^2 + (\phi_2 - \phi_3)L^3 + \phi_3 L^4] y_t \\ = (1 - \theta_1 L)\varepsilon_t, \end{aligned}$$

$$\begin{aligned} y_t - (1 + \phi_1)y_{t-1} + (\phi_1 - \phi_2)y_{t-2} + (\phi_2 - \phi_3)y_{t-3} \\ + \phi_3 y_{t-4} = \varepsilon_t - \theta_1 \varepsilon_{t-1}. \end{aligned}$$

$$\begin{aligned} y_t = (1 + \phi_1)y_{t-1} - (\phi_1 - \phi_2)y_{t-2} - (\phi_2 - \phi_3)y_{t-3} \\ - \phi_3 y_{t-4} + \varepsilon_t - \theta_1 \varepsilon_{t-1}. \end{aligned}$$

Point forecasts ($S=1$)

$$y_t = (1 + \phi_1)y_{t-1} - (\phi_1 - \phi_2)y_{t-2} - (\phi_2 - \phi_3)y_{t-3} \\ - \phi_3y_{t-4} + \varepsilon_t - \theta_1\varepsilon_{t-1}.$$

ARIMA(3, 1, 1) forecasts: Step 2

$$y_{T+1} = (1 + \phi_1)y_T - (\phi_1 - \phi_2)y_{T-1} - (\phi_2 - \phi_3)y_{T-2} \\ - \phi_3y_{T-3} + \varepsilon_{T+1} - \theta_1\varepsilon_T.$$

ARIMA(3, 1, 1) forecasts: Step 3

$$\hat{y}_{T+1|T} = (1 + \phi_1)y_T - (\phi_1 - \phi_2)y_{T-1} - (\phi_2 - \phi_3)y_{T-2} \\ - \phi_3y_{T-3} - \theta_1\varepsilon_T.$$

Point forecasts (S=2)

$$y_t = (1 + \phi_1)y_{t-1} - (\phi_1 - \phi_2)y_{t-2} - (\phi_2 - \phi_3)y_{t-3} \\ - \phi_3y_{t-4} + \varepsilon_t - \theta_1\varepsilon_{t-1}.$$

ARIMA(3, 1, 1) forecasts: Step 2

$$y_{T+2} = (1 + \phi_1)y_{T+1} - (\phi_1 - \phi_2)y_T - (\phi_2 - \phi_3)y_{T-1} \\ - \phi_3y_{T-2} + \varepsilon_{T+2} - \theta_1\varepsilon_{T+1}.$$

ARIMA(3, 1, 1) forecasts: Step 3

$$\hat{y}_{T+2|T} = (1 + \phi_1)\hat{y}_{T+1|T} - (\phi_1 - \phi_2)y_T - (\phi_2 - \phi_3)y_{T-1} \\ - \phi_3y_{T-2}.$$

Prediction Intervals

- Assuming that $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$, the 95% Prediction Intervals can be obtained as:

$$CI_{95\%}(y_{T+S}) = y_{T+S|T} \pm 1.96\sigma_{\varepsilon_{T+S|T}}$$

- $\sigma_{\varepsilon_{T+1|T}}^2 = \hat{\sigma}^2$
- Multi-step prediction intervals for $ARIMA(0, 0, q)$:

$$y_t = \varepsilon_t - \sum_{i=1}^q \theta_i \varepsilon_{t-i}$$

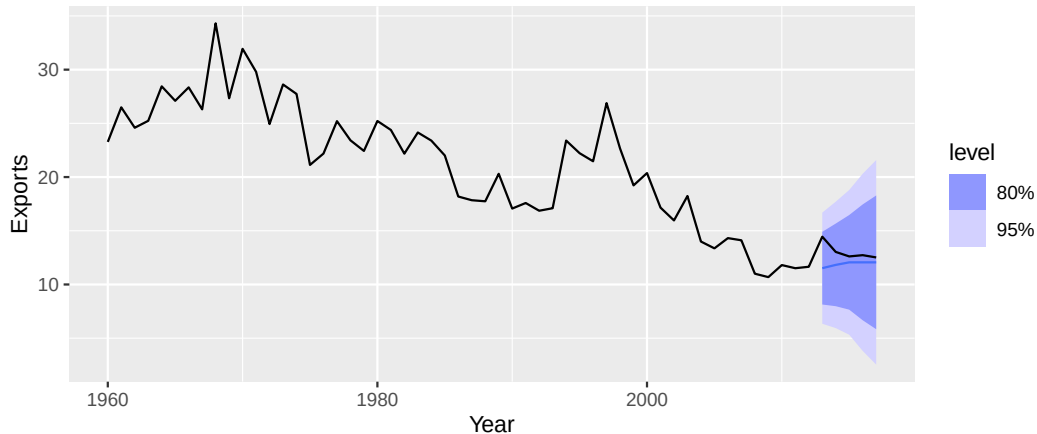
$$\sigma_{\varepsilon_{T+S|T}}^2 = \sigma^2 \left[1 + \sum_{i=1}^{S-1} \theta_i \right], \text{ for } S = 2, 3, \dots$$

Other models? Use invertibility result!

Forecasting five periods ahead

```
caf_fit %>%  
  forecast(h=5) %>%  
  filter(.model=='arima013') %>%  
  autoplot(global_economy)
```

Forecasting five periods ahead



In-sample vs Out-of-sample Forecasts

- In any forecasting exercise, it is important to distinguish between:
 - Estimation sample (training sample)
 - Forecast sample (test sample)
- Furthermore in any forecasting exercise, it is important to understand if we want to evaluate the forecasting accuracy with:
 - In-sample forecasts
 - Out-of-sample forecasts

In-sample vs Out-of-sample Forecasts

- The following example helps to understand the difference between these concepts.
- Example: suppose that we want to forecast US Industrial Production Index and we have data from 1960M1 until 2019M11.
 - Estimation sample: use data from 1960M1 to 2019M5 to estimate the model
 - Forecast sample: to obtain only out-of-sample forecasts we can make predictions starting from 2019M6 until 2012M11.
 - We can obtain in-sample forecasts with predictions from 1960M1 until 2019M5
- Out-of-sample forecasts are much more interesting than in-sample forecasts to evaluate the model prediction accuracy.
- Suppose that the IPI forecast for the next month is 109.4 but the realized value turned out to be 108.7. Is this a good or a bad forecast?

Forecasting accuracy

- With only one prediction and actual value it is not possible to evaluate the prediction accuracy of a given model.
- In general, we compute out-of-sample forecasts and the corresponding out-of-sample prediction errors.

Forecasting accuracy

We then aggregate this information into some sort of indicator that can be used for comparison purposes:

- $RMSE = \sqrt{\frac{\sum_{t=T+1}^{T+s} (y_t - \hat{y}_t)^2}{s}}$
- $MAE = \sum_{t=T+1}^{T+s} \frac{|y_t - \hat{y}_t|}{s}$
- $MAPE = 100 \sum_{t=T+1}^{T+s} \frac{\frac{|y_t - \hat{y}_t|}{y_t}}{s}$
- Theil inequality coefficient $U = \frac{\sqrt{\frac{\sum_{t=T+1}^{T+s} (y_t - \hat{y}_t)^2}{s}}}{\sqrt{\frac{\sum_{t=T+1}^{T+s} \hat{y}_t^2}{s}} + \sqrt{\frac{\sum_{t=T+1}^{T+s} y_t^2}{s}}}$

Forecasting accuracy

```
#Training data
caf_train <- global_economy %>%
  filter(Code == "CAF") %>%
  filter(Year <= 2012)

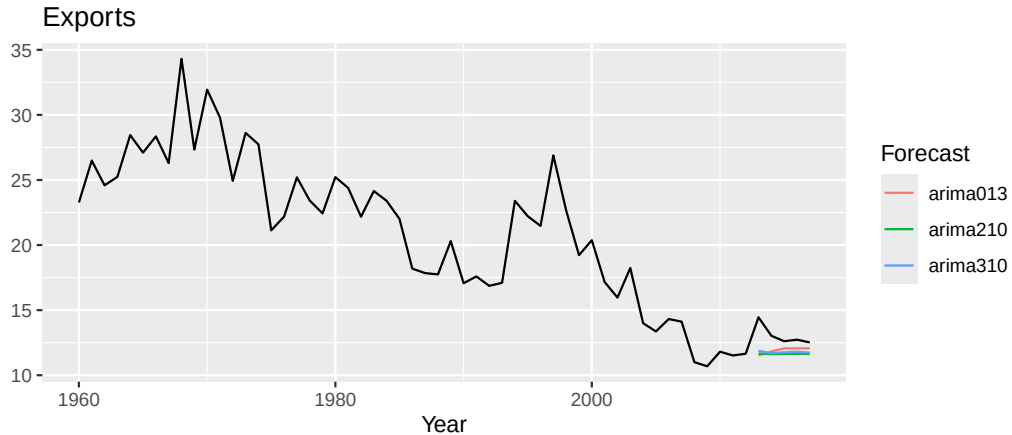
#Test data
caf_test <- global_economy %>%
  filter(Code == "CAF") %>%
  filter(Year > 2012)

#Fit models in training data
caf_fit <- caf_train %>%
  model(arima210 = ARIMA(Exports ~ pdq(2,1,0)),
        arima013 = ARIMA(Exports ~ pdq(0,1,3)),
        arima310 = ARIMA(Exports ~ pdq(3,1,0)))

#Forecast in test data
caf_fc <- caf_fit %>%
  forecast(caf_test)

#Plot results
caf_fc %>%
  autoplot(bind_rows(caf_train, caf_test),
            level = NULL) +
  labs(y = '',
        title = 'Exports') +
  guides(colour = guide_legend(title = "Forecast"))
```

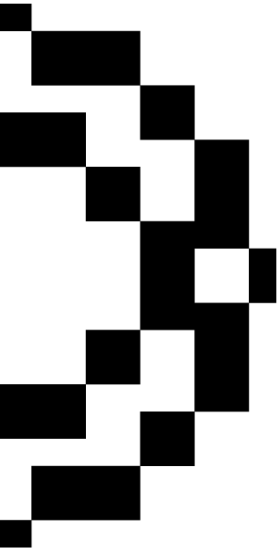
Forecasting accuracy



Forecasting accuracy

```
accuracy(caf_fc, global_economy)
```

```
## # A tibble: 3 x 11
##   .model Country      .type  ME  RMSE  MAE  MPE  MAPE  MASE  RMSSE  ACF1
##   <chr>    <fct>      <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 arima013 Central Africa~ Test  1.16  1.48  1.16  8.57  8.57  0.505  0.495  0.165
## 2 arima210 Central Africa~ Test  1.43  1.59  1.43 10.7  10.7  0.620  0.530  0.132
## 3 arima310 Central Africa~ Test  1.30  1.46  1.30  9.68  9.68  0.562  0.486  0.176
```



Introduction to SARIMA Models

Seasonal ARIMA models

- So far, we have restricted our attention to non-seasonal data and non-seasonal ARIMA models. However, ARIMA models are also capable of modelling a wide range of seasonal data.
- A seasonal ARIMA model is formed by including additional seasonal terms in the ARIMA models we have seen so far. It is written as follows:

$$ARIMA \quad \underbrace{(p, d, q)}_{\text{Non-seasonal part}} \quad \underbrace{(P, D, Q)_m}_{\text{Seasonal part}}$$

where m is the seasonal period (e.g., number of observations per year)

ACF/PACF

- The seasonal part of an AR or MA model will be seen in the seasonal lags of the PACF and ACF. For example, an $ARIMA(0,0,0)(0,0,1)_{12}$ model will show:
 - a spike at lag 12 in the ACF but no other significant spikes;
 - exponential decay in the seasonal lags of the PACF (i.e., at lags 12, 24, 36, ...).
- Similarly, an $ARIMA(0,0,0)(1,0,0)_{12}$ model will show:
 - exponential decay in the seasonal lags of the ACF;
 - a single significant spike at lag 12 in the PACF.
- In considering the appropriate seasonal orders for a seasonal ARIMA model, restrict attention to the seasonal lags.

Example

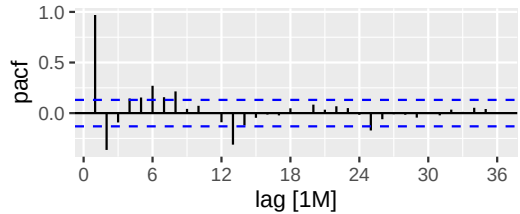
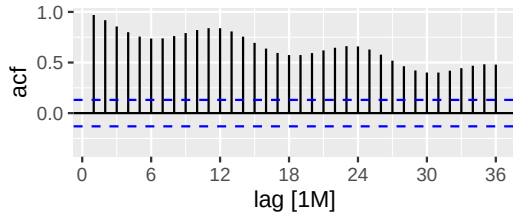
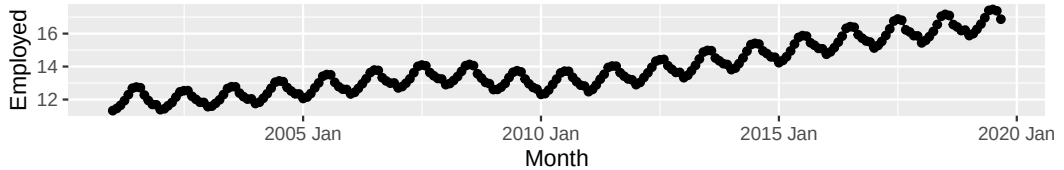
We will describe seasonal ARIMA modelling using monthly US employment data for leisure and hospitality jobs from January 2000 to September 2019.

Example

```
leisure <- us_employment %>%  
  filter(Title == "Leisure and Hospitality",  
         year(Month) > 2000) %>%  
  mutate(Employed = Employed/1000) %>%  
  select(Month, Employed)  
gg_tsddisplay(leisure, Employed, plot_type='partial', lag=36) +  
  labs(title = "US employment: leisure and hospitality",  
       y="Number of people (millions)")
```

Example

US employment: leisure and hospitality



Example

Apparently, by analysing the plot, we have a strong seasonality and a nonlinear trend. The first step is to test the presence of a unit root.

Example

```
library(urca)
summary(ur.df(leisure$Employed, type=c('trend'),
              selectlags=24))
```

Example

Value of test-statistic is: -1.845 2.5666 2.6289

Critical values for test statistics:

	1pct	5pct	10pct
tau3	-3.99	-3.43	-3.13
phi2	6.22	4.75	4.07
phi3	8.43	6.49	5.47

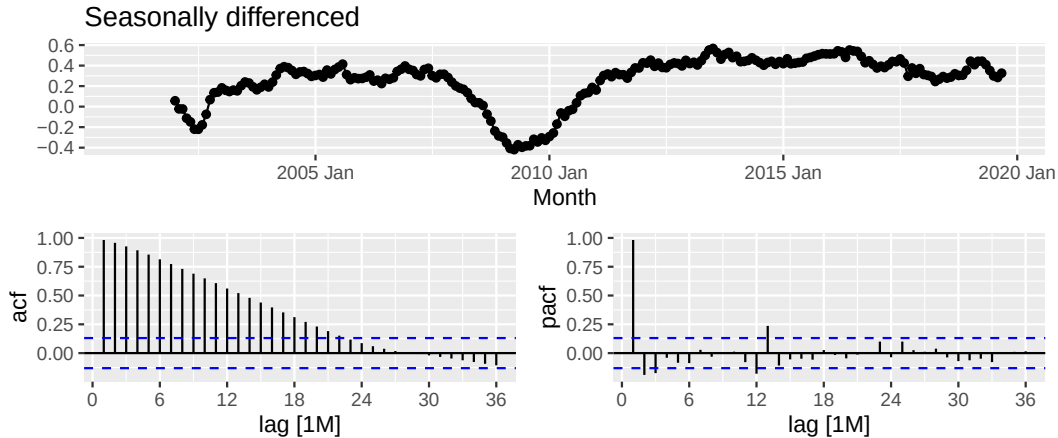
Example

We don't reject the null hypothesis, hence we will seasonally difference the series.

Example

```
leisure %>%  
  gg_tsdisplay(difference(Employed, 12),  
               plot_type='partial', lag=36) +  
  labs(title="Seasonally differenced", y="")
```

Example



Example

Taking the seasonal differences is not enough, we need to test once more for the presence of unit root.

Example

```
leisure <- leisure %>%  
  mutate(d12_Employed = difference(Employed, 12))  
library(urca)  
summary(ur.df(na.omit(leisure$d12_Employed), type=c('drift'),  
  lags=24))
```

Example

Value of test-statistic is: -1.7358 1.5409

Critical values for test statistics:

	1pct	5pct	10pct
tau2	-3.46	-2.88	-2.57
phi1	6.52	4.63	3.81

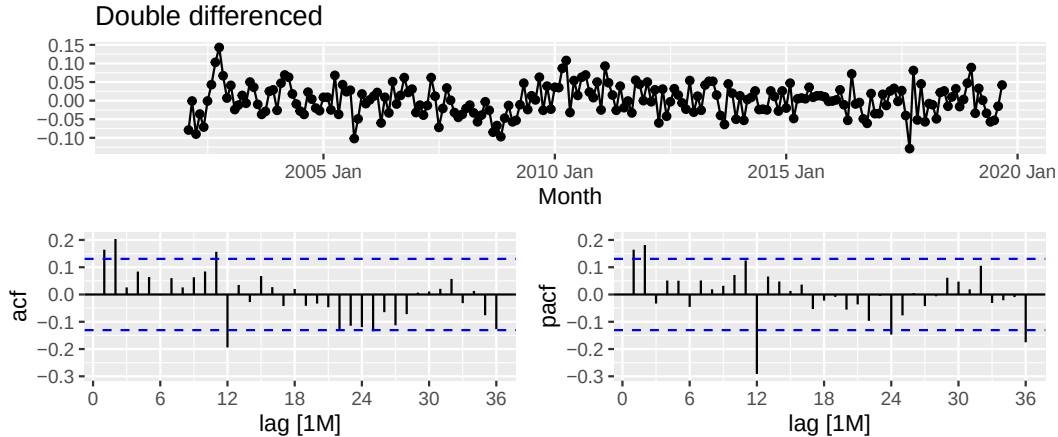
Example

We don't reject the null hypothesis, so we take a further first difference.

Example

```
leisure %>%  
  gg_tsdisplay(difference(Employed, 12) %>% difference(),  
               plot_type='partial', lag=36) +  
  labs(title = "Double differenced", y="")
```

Example



Example

Let's test once more for the presence of unit root on the double difference series.

Example

```
leisure <- leisure %>%  
  mutate(dd_Employed = difference(d12_Employed, 1))  
library(urca)  
summary(ur.df(na.omit(leisure$dd_Employed), type=c('none'),  
  lags=23))
```

Example

Value of test-statistic is: -3.6474

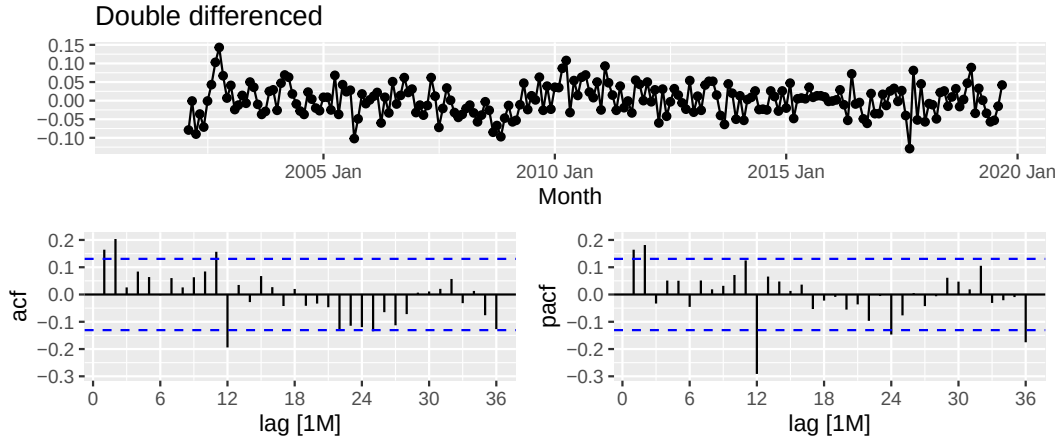
Critical values for test statistics:

	1pct	5pct	10pct
tau1	-2.58	-1.95	-1.62

Example

We reject the null hypothesis, so we don't take further differences. Thus, we will analyse the ACF and PACF of the double differenced series to select the order of the model.

Example



Example

- We start by selecting an $ARIMA(2, 1, 0)(0, 1, 1)_{12}$ model - using the PACF to select the non-seasonal part of the model and the ACF to select the seasonal part of the model.
- We can extend this to an $ARIMA(1, 1, 0)(0, 1, 1)_{12}$ model, as a particular case from this general model.

Example

```
fit <- leisure %>%
  model(
    arima110011 = ARIMA(Employed ~ pdq(1,1,0) + PDQ(0,1,1)),
    arima210011 = ARIMA(Employed ~ pdq(2,1,0) + PDQ(0,1,1)),
  )
fit %>% pivot_longer(everything(), names_to = "Model name",
  values_to = "Orders")

## # A mable: 2 x 2
## # Key:   Model name [2]
##   `Model name`           Orders
##   <chr>                <model>
## 1 arima110011 <ARIMA(1,1,0)(0,1,1)[12]>
## 2 arima210011 <ARIMA(2,1,0)(0,1,1)[12]>

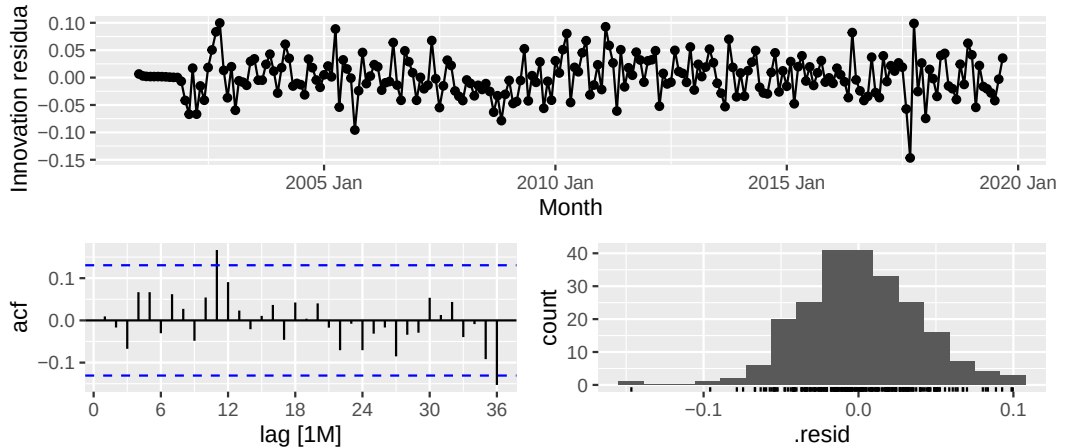
glance(fit) %>% arrange(AICc) %>% select(.model:BIC)

## # A tibble: 2 x 6
##   .model      sigma2 log_lik   AIC  AICc   BIC
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <dbl>
## 1 arima210011 0.00145   392. -776. -776. -763.
## 2 arima110011 0.00150   388. -770. -770. -760.
```

Example

We select the model $ARIMA(2, 1, 0)(0, 1, 1)_{12}$ according to the information criteria. Checking the residuals for this estimation...

Example



Example

Some significant spikes appear in the ACF plot. Next step is to test for the presence of autocorrelation in the residuals through the Ljung-Box and Box-Pierce tests...

Example

```
augment(fit) %>%  
  filter(.model=='arima210011') %>%  
  features(.innov, ljung_box, lag = 36)
```

```
## # A tibble: 1 x 3  
##   .model      lb_stat lb_pvalue  
##   <chr>      <dbl>    <dbl>  
## 1 arima210011  32.0      0.662
```

Example

```
augment(fit) %>%  
  filter(.model=='arima210011') %>%  
  features(.innov, box_pierce, lag = 36)
```

```
## # A tibble: 1 x 3  
##   .model      bp_stat bp_pvalue  
##   <chr>      <dbl>    <dbl>  
## 1 arima210011  28.8      0.799
```

Example

The p -values for these tests are all greater than 5%, so we don't reject the null hypothesis, which means we don't have statistical evidence of the presence of autocorrelation in the residuals. We can move on to the final step: forecasting the series!

Example

```
fabletools::forecast(fit, h=36) %>%  
  filter(.model=='arima210011') %>%  
  autoplot(leisure) +  
  labs(title = "US employment: leisure and hospitality",  
        y="Number of people (millions)")
```

Example

